

Backend Development Task for Interns

Objective:

Develop the **desktop and responsive (mobile view)** functionality for the eCommerce web design template provided in the Figma file: [Ecommerce Web Design](#). This task focuses on backend development while integrating the design template with dynamic and responsive capabilities.

Tasks Overview:

Each week is structured to guide you through backend development, integrating frontend designs with dynamic content, and ensuring a fully responsive layout.

Week 1: Project Setup and Static Backend Integration

1. **Goal:** Set up the project environment and create static endpoints to serve the frontend templates.
2. **Tasks:**
 - Set up the backend server using **Node.js** and **Express.js**.
 - Organize the project directory with folders for routes, views, public (CSS/JS), and assets.
 - Create static HTML/CSS files for:
 - Home Page
 - Product Listing Page
 - Product Details Page
 - Serve these static pages through Express routes (`/`, `/products`, `/products/:id`).
 - Ensure mobile responsiveness of all pages using media queries in the provided CSS files.
3. **Deliverables:**
 - Functional Node.js server with routes serving static pages.
 - Desktop and mobile view consistency using responsive design techniques.
 - Push code to a GitHub repository.

Week 2: Database Integration and Dynamic Content

1. **Goal:** Connect to a database and render dynamic content on the frontend.

2. **Tasks:**

Set up a **MongoDB database** (or Firebase Realtime Database if preferred).

- Create a collection for **Products** with attributes like **id**, **name**, **price**, **category**, **image**, **description**, and **stock**.
- Populate the database with sample product data.
- Use **EJS** (or another templating engine) to dynamically render:
 - Home Page: Display featured products from the database.
 - Product Listing Page: Fetch and display all products from the database in a grid layout.
 - Product Details Page: Fetch and display detailed information for a single product.
- Implement a **search bar** (functional, server-side) to filter products by name or category.

3. **Deliverables:**

Fully integrated backend with dynamic data rendering.

- Responsive design for all pages (desktop and mobile).
- Code committed to GitHub repository.

Week 3: Backend Features and Final Integration

1. **Goal:** Add backend features and finalize the project for deployment.

2. **Tasks:**

Implement **user authentication** using **JWT** or **Firebase Auth**.

- Create login and signup pages.
- Restrict access to certain pages (e.g., adding products, admin actions).
- Add a feature to **add products** using a form and save them to the database.
- Implement **pagination** for the Product Listing Page to handle large datasets.
- Deploy the project to a platform like **Render**, **Heroku**, or **Vercel**.

3. **Deliverables:**

Authentication system implemented and tested.

- Pagination and add product functionality working as expected.
- Deployed project with live URL.
- Final code uploaded to the GitHub repository.

Tools and Technologies:

- **Node.js** and **Express.js** for backend development.
- **MongoDB** (or Firebase) for the database.
- **EJS** (or any templating engine) for dynamic content rendering.
- **HTML5/CSS3/JavaScript** for frontend integration.
 - **GitHub** for version control and code submission.
- OR YOU CAN USE TECHNOLOGY STACK OF YOUR OWN CHOICE

Submission Instructions:

1. Create a GitHub repository named **ecommerce-backend-design**.
2. Push your code weekly based on deliverables.
3. Share the repository link and deployed URL (Week 3) with your for review.

Evaluation Criteria:

1. **Functionality:**
 - All routes are functional and serve dynamic data.
 - Responsive design on both desktop and mobile.
2. **Code Quality:**
 - Organized folder structure and readable code.
 - Proper error handling in backend routes.
3. **Database Design:**
 - Efficient schema for managing products.
 - Secure and functional authentication.
4. **Timely Submission:**
 - Complete weekly tasks on time and push to GitHub.

Deadline

- **Final Submission Date: 5th June, 2026**
- **You will have to submit all of these tasks collectively on 5th June, 2026**

Good luck! Ensure the integration is seamless and both desktop and mobile views are responsive.